

DP Class - 6

Special class

Dp $\rightarrow$ inc/exc ✓

$\rightarrow$ with strings

$\rightarrow$ with BS

$\rightarrow$ all possibility

$\rightarrow$ tree

$\rightarrow$ merge interval

$\rightarrow$ pattern matching

~~Range~~ LIS

→ wildcard pattern matching

str →

"~~l~~~~a~~~~b~~~~b~~~~a~~~~y~~"

pattern →

"~~l~~~~a~~?~~*~~~~*~~~~*~~~~*~~"

} True

? → any single character

{ a, b, c, ..., z }

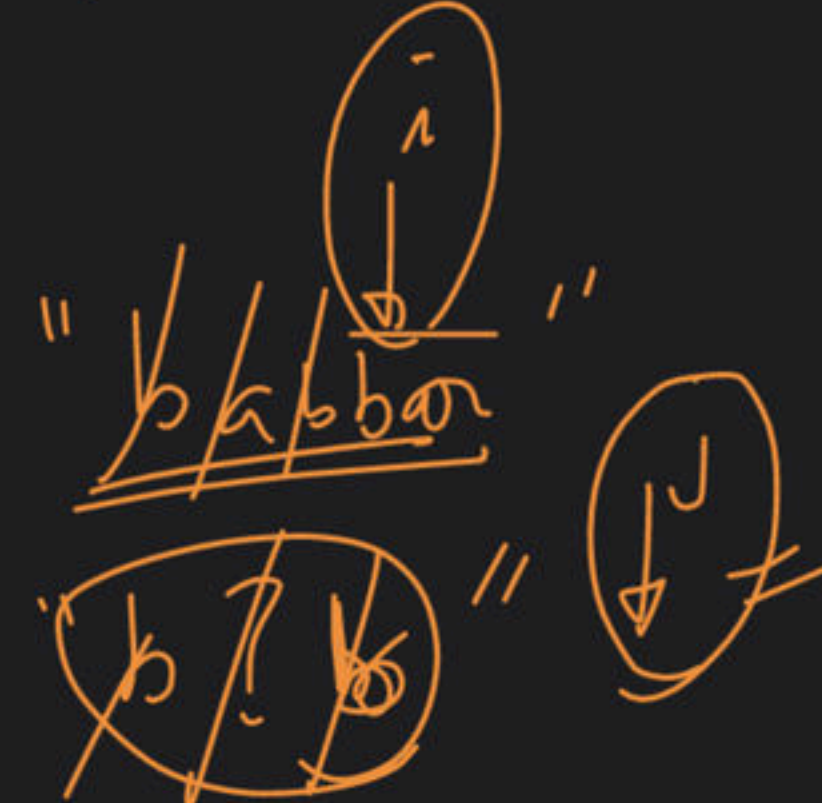
★ → matches any character
any no. of character

$s \rightarrow$ "babbar"
 $p \rightarrow$ "b*bb*


(I)
 $i == s.length() \ \&\& \ j == p.length()$
 true

?

★

$s =$ "babbar"
 $p =$ "b*bb*


(II)
 $i == p.length()$
 false

s →

~~babbar~~

i

p →

~~babbaring~~

s →

~~babbar~~

p →

~~babbar~~

babbar

~~s~~ → false

IV

logic

Match

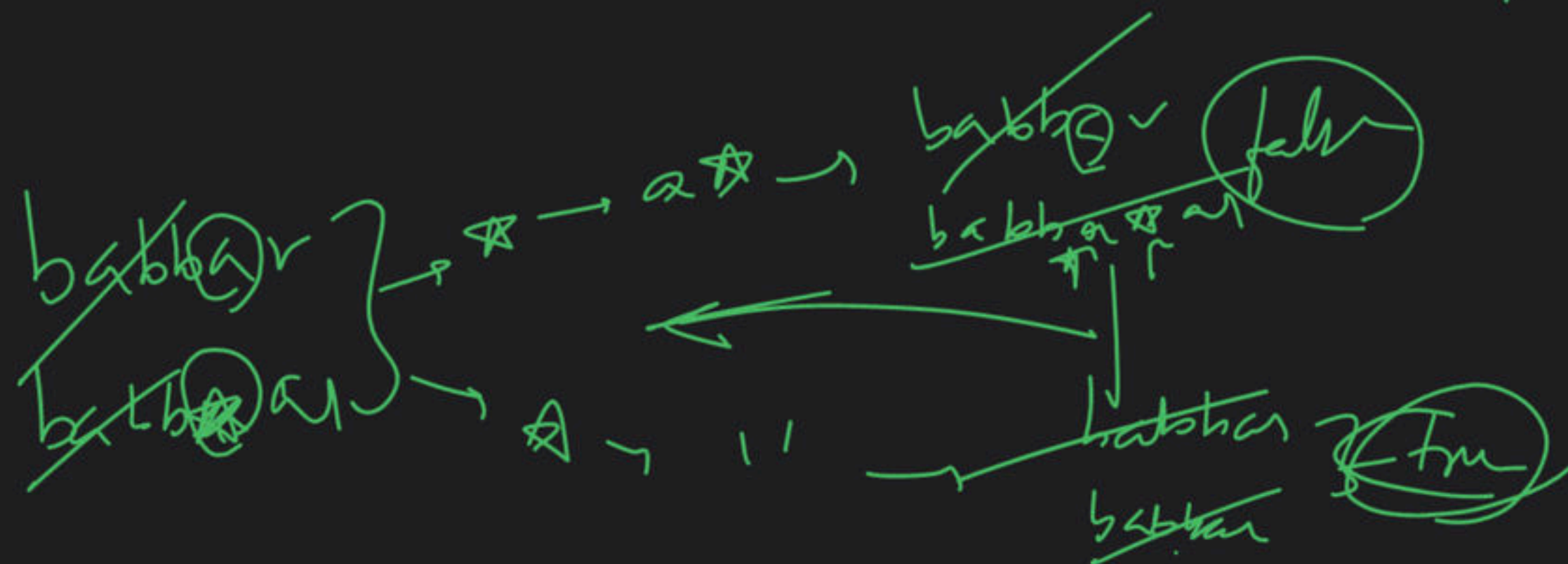
no match

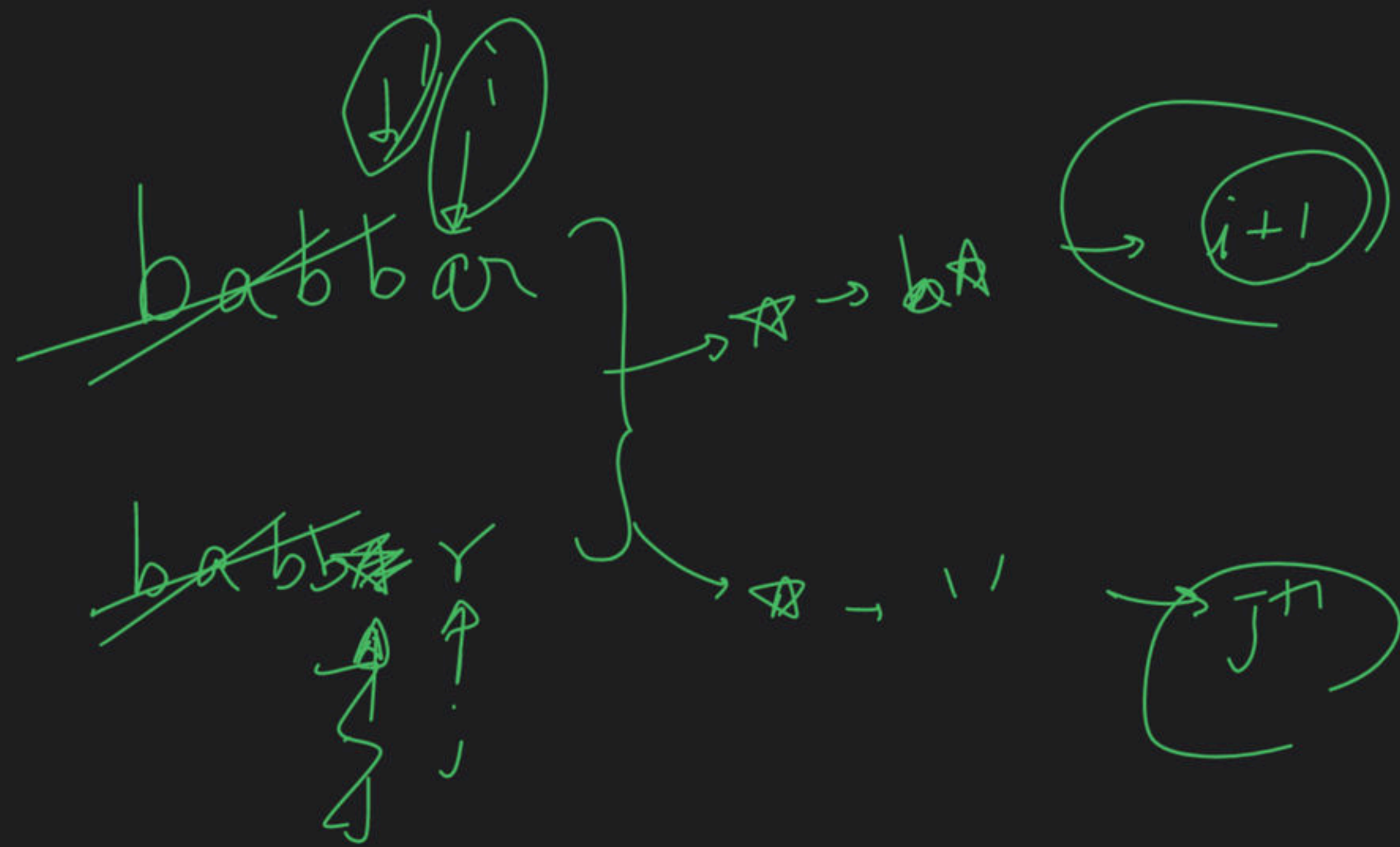
$i == s.length() \wedge j < p.length()$

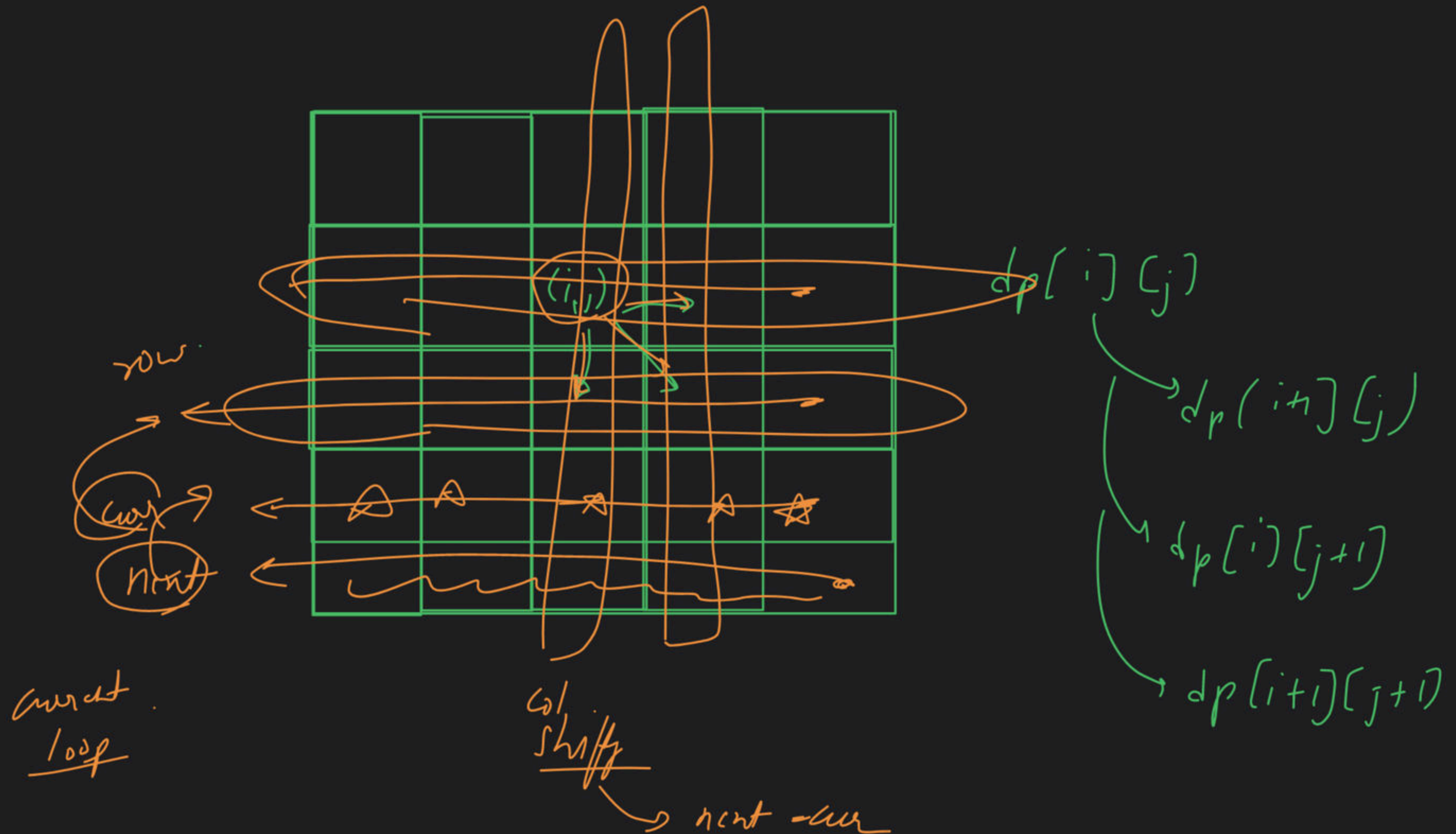
→ same hache hne * h

↳ T

→ nah hai → false

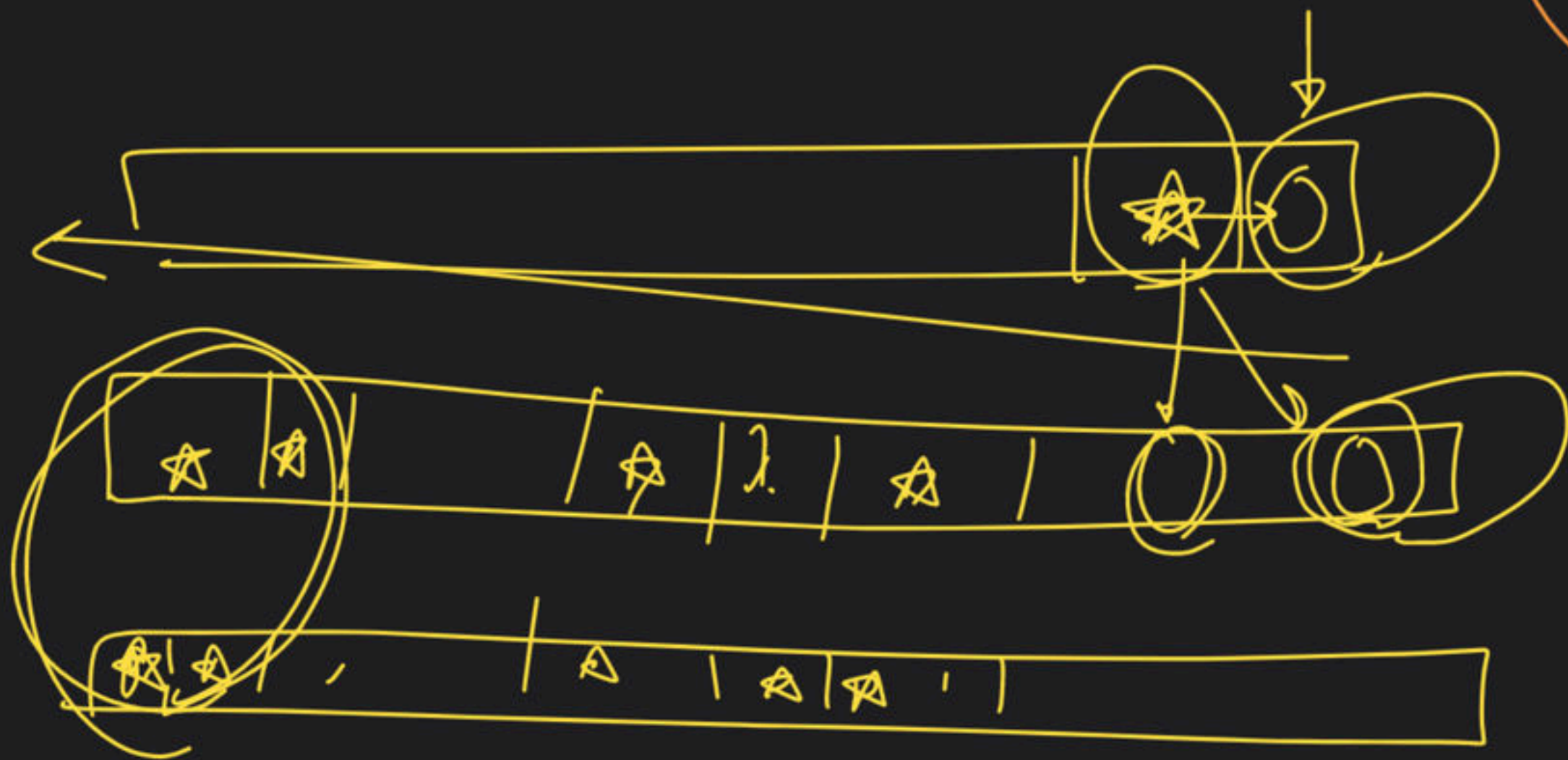








$curr[j]$ $\rightarrow$ $wn[j+1]$
 $\rightarrow$ $hint[j]$
 $\rightarrow$ $next[j+1]$





"bab~~ba~~"

True

"bab~~ba~~"



.ba
a~~a~~
a~~a~~
|a

→ bab~~ba~~r

} false

→ bab~~ba~~~~a~~

... b, bb, bbb, bbb ...

wildcard

~~b~~ ~~a~~ ~~b~~ ~~b~~ ~~d~~ ~~r~~ } d
~~b~~ ~~a~~ ~~b~~ ~~*~~ ~~*~~ ~~r~~ } d

I
 $i := s.\text{len}() \ \& \ j := p.\text{len}()$
 true

III
 $i == s.\text{len}()$
 true
 false

II
 $j^2 = p.\text{len}()$
 false

i $i+1$
~~b~~a~~b~~a~~r~~
~~b~~a~~b~~a~~r~~

i
~~b~~a~~b~~a~~r~~
 i j T
~~b~~a~~b~~a~~r~~
 $i-1$ $j-1$
 $T \rightarrow i$
 $T \rightarrow j$
 $T \rightarrow b$
 babbar (only)
 babar
 babbar
 babbar

$\star$ $\rightarrow$ $\bigcirc$ $\rightarrow$ Cur Match
 $\rightarrow j+2$

$\star$ $\rightarrow$ $\bigcirc$ $\rightarrow$ prev $\star$
 $\rightarrow i+1$

T $\rightarrow$ $\star$ $\rightarrow$ $\star$ $\rightarrow$ $\star$
 $\rightarrow i$ $\rightarrow b$
~~babbar~~
~~babbar~~
 babbar
 babbar

~~ba~~bbbar } $\xrightarrow{\star}$ ' '

~~ba~~bbbar }

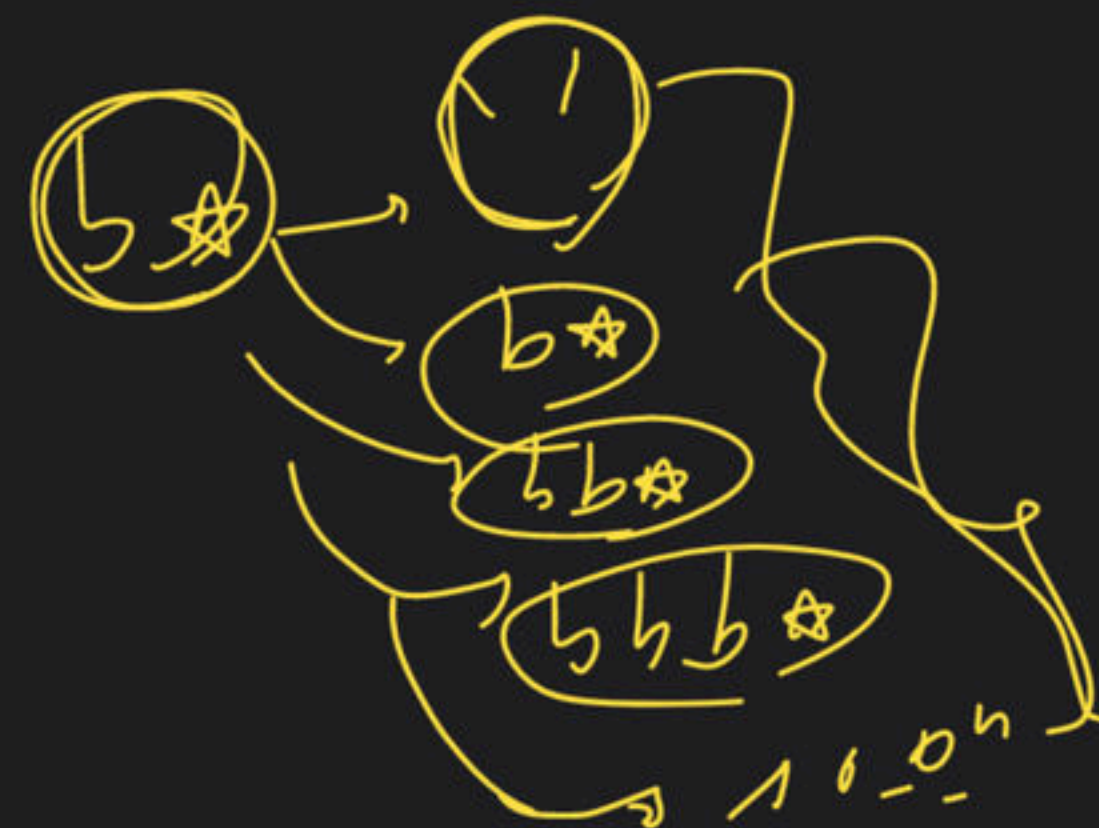
bbar
bmar

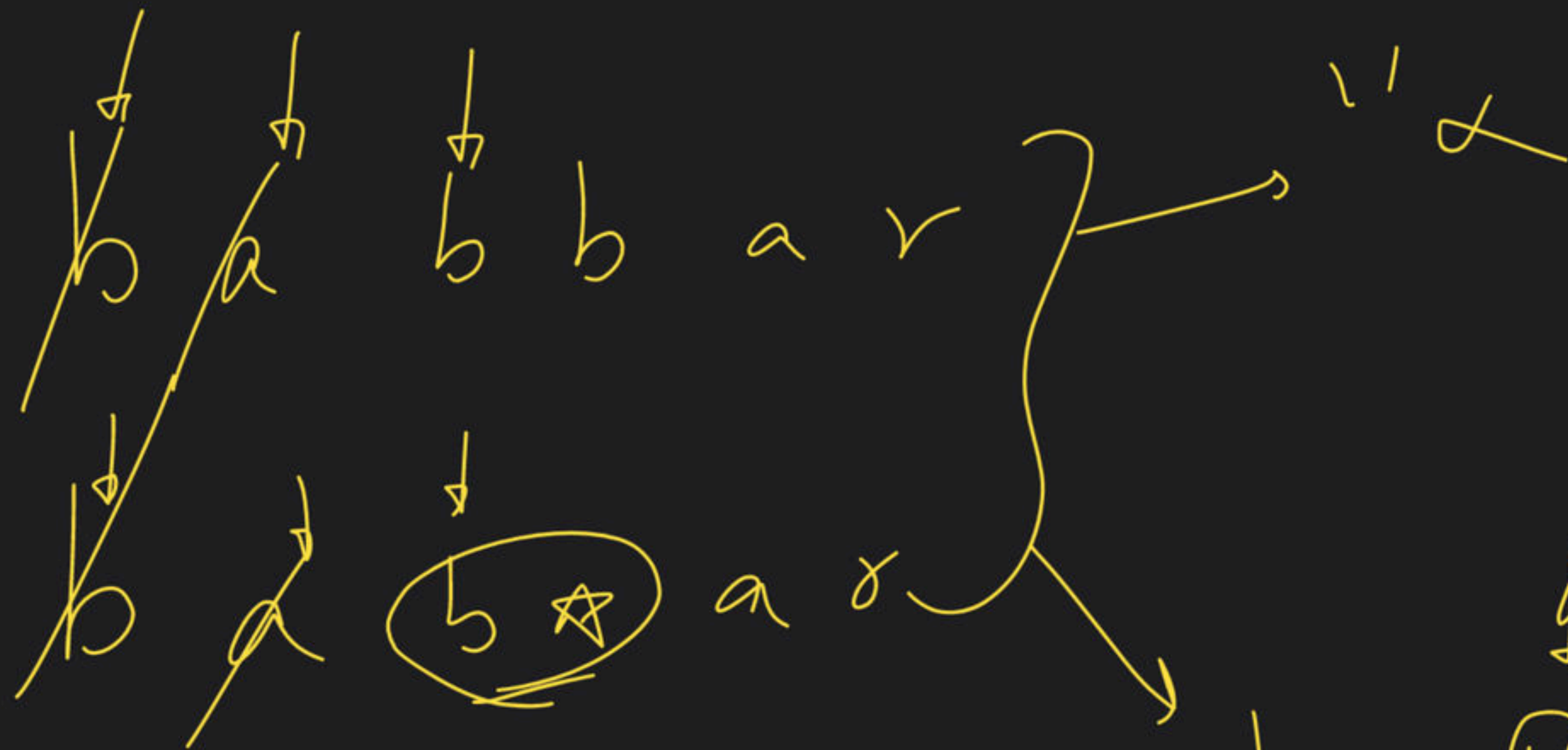
~~b~~ ~~a~~ ~~b~~ b a r

~~b~~ ~~a~~ (b ~~a~~ r)

b a (b b a r)
b a ~~x~~ ~~x~~ (a ~~x~~)
 j j+2

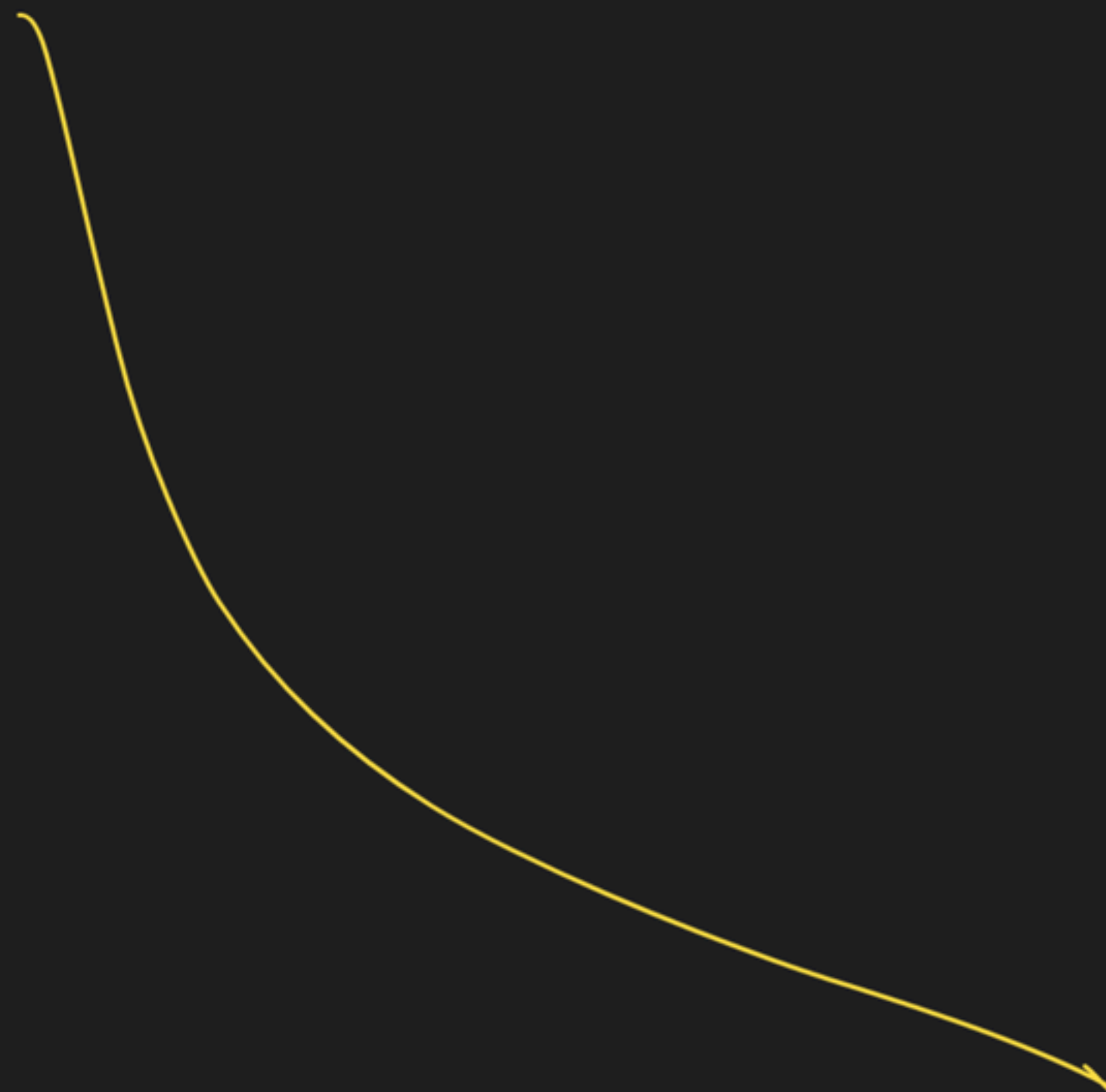
empty $\rightarrow f(i, j+2)$

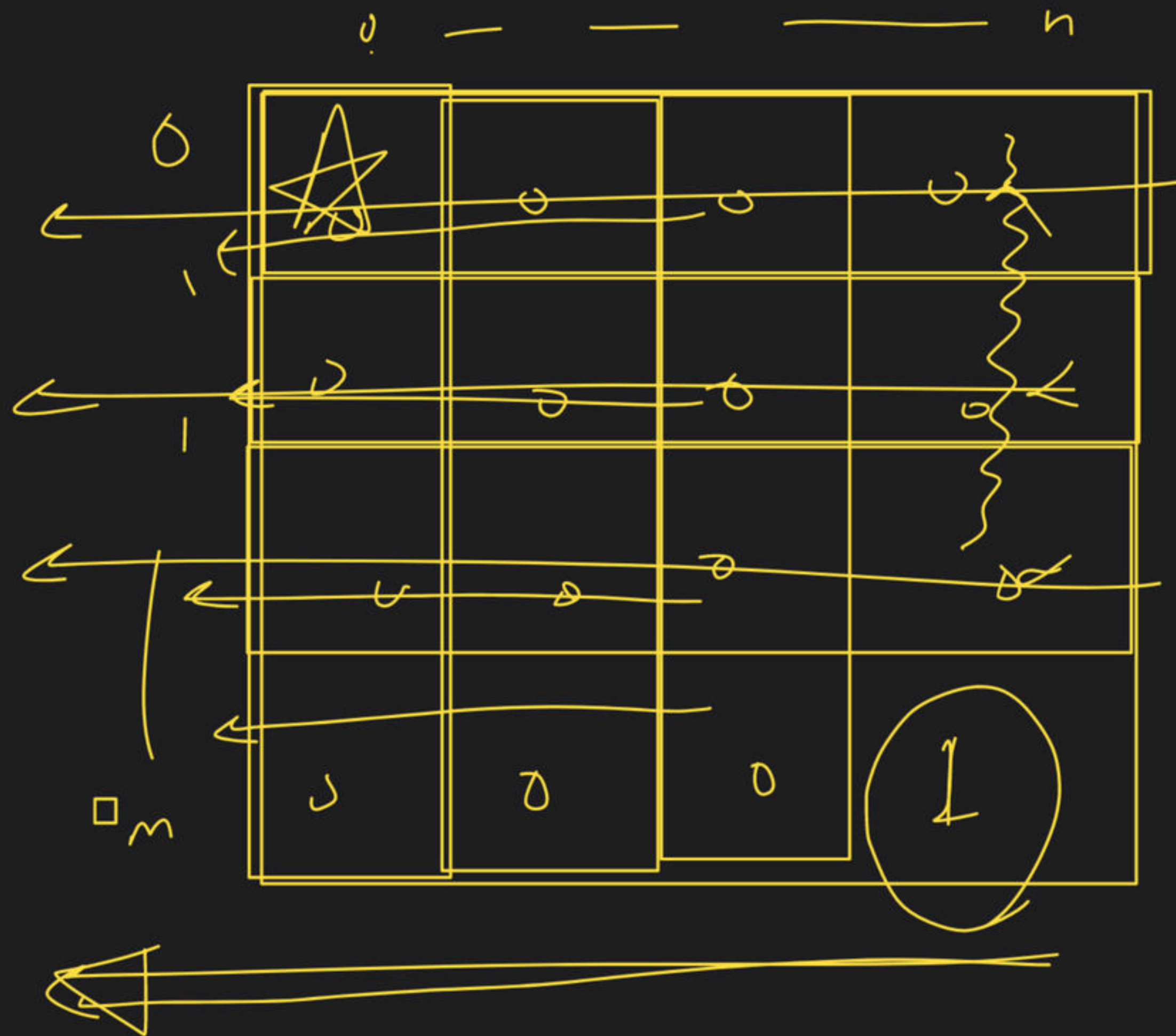




Answer Match & l f (i+1, j)







$dp[m][p]$

$n-1$

• → single ch



$\downarrow \downarrow \downarrow$
~~b~~ ~~a~~ b b a r
 $\downarrow$
~~b~~ ~~a~~ (b $\star$) a r
 $\downarrow$
~~b~~ ~~a~~ (b b $\star$)

cur match
 at (17)

$\downarrow$
 b a (b b a r) $\rightarrow$ R.C
~~b~~ ~~a~~ $\bar{x}$ $\bar{x}$ (a r)
 (7) $\uparrow$ (J+2)

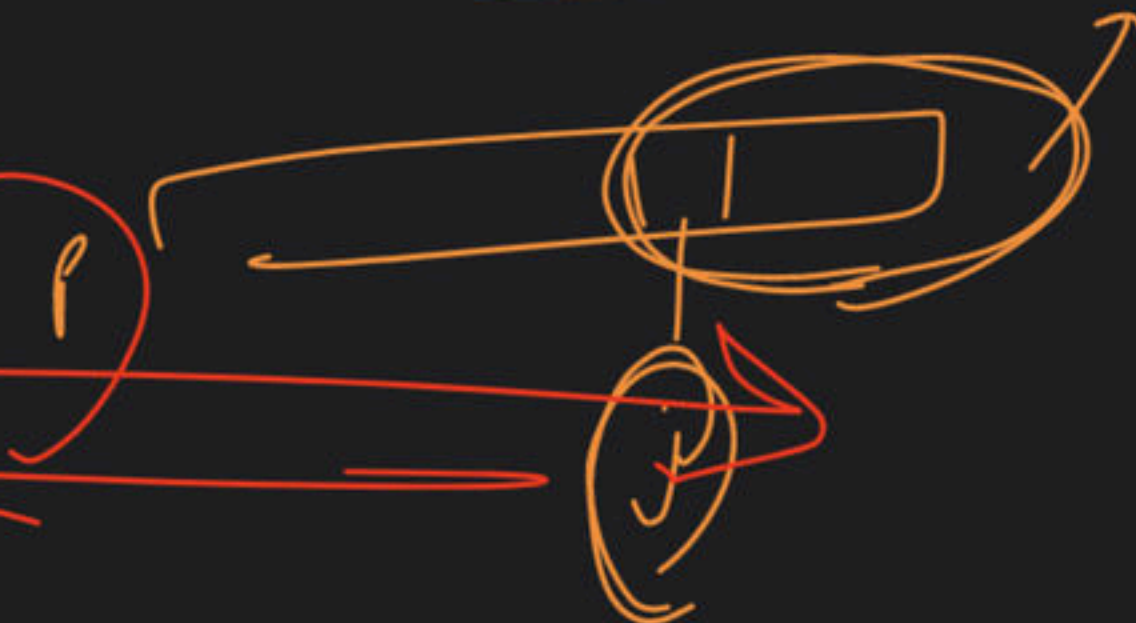
b a (b b a r) (17)
 b a (b b a r) (17)
 b a (b b a r) (17)

$\delta \rightarrow \cdot$
 $p-1$ q $\rightarrow$ Time

	0	1	...	n
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
...				
m-1	0	0	0	0
m	0	0	0	0

$dp[m][n]$

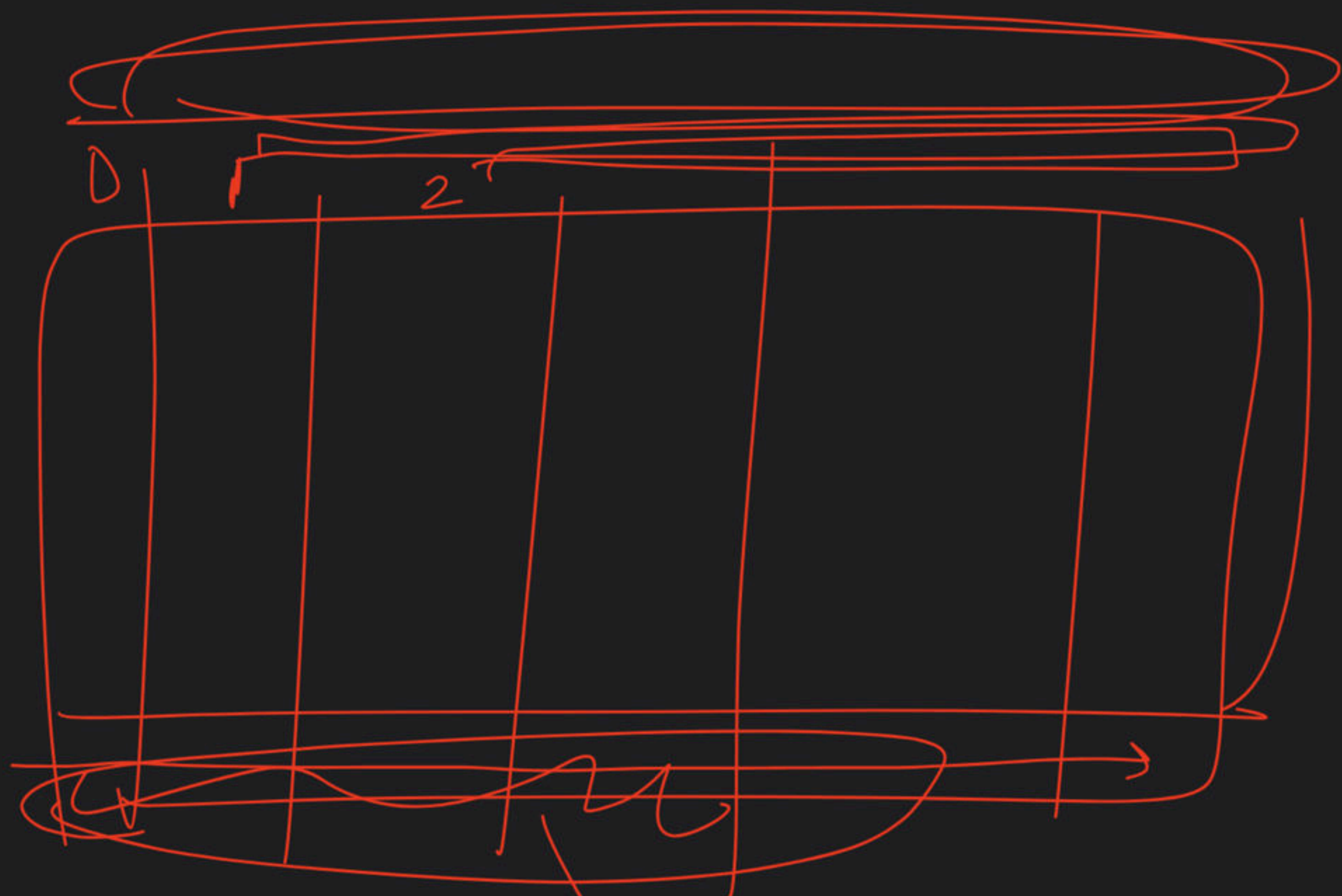
$dp[i][j] \rightarrow$



m

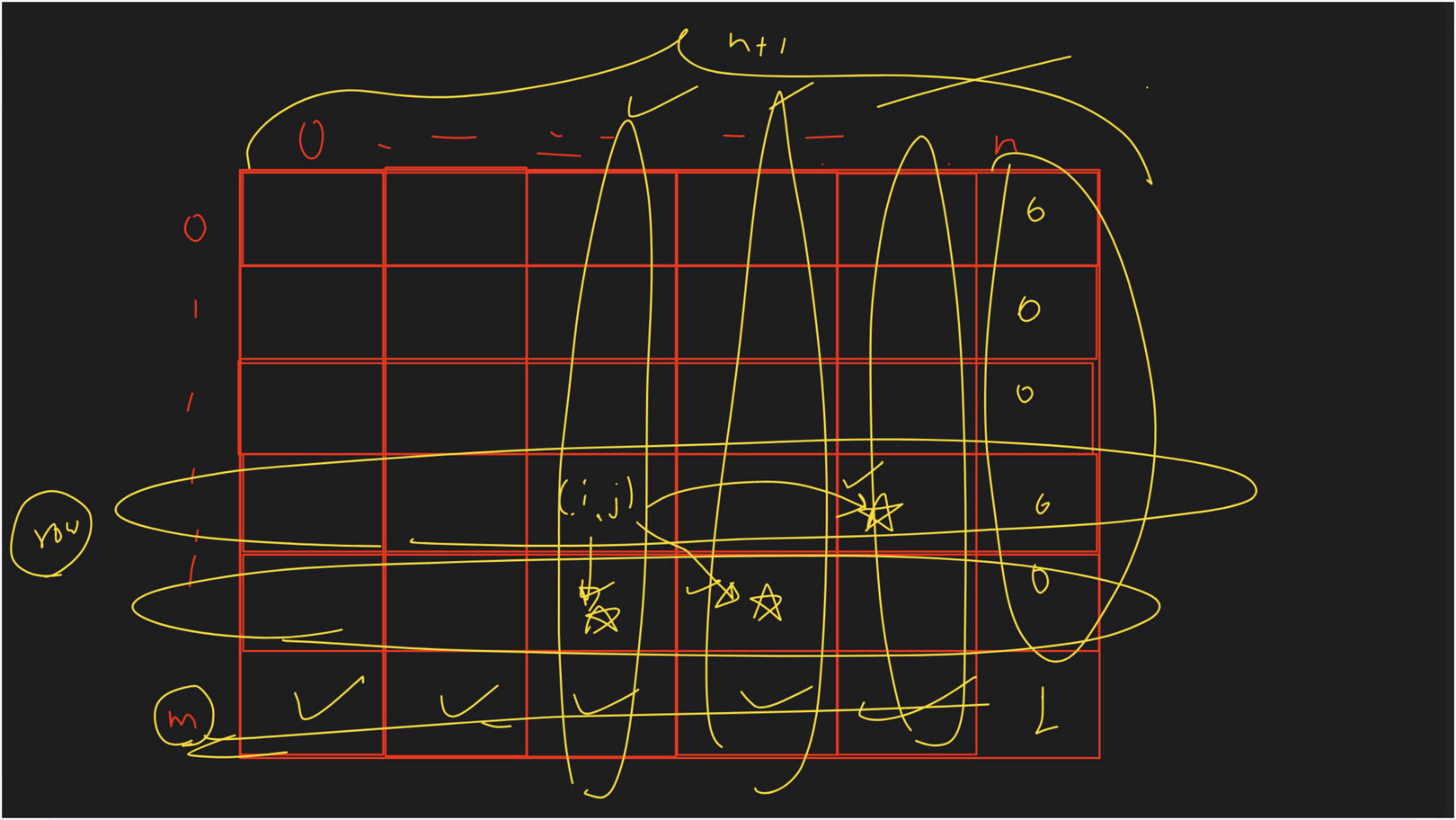
n

m



d -> ()
p

?



CW

0
0
0
0
0

curr

0	0	0	0	0	0
---	---	---	---	---	---

next

0	0	0	0	0	1
---	---	---	---	---	---

last row

$l == m$

curr

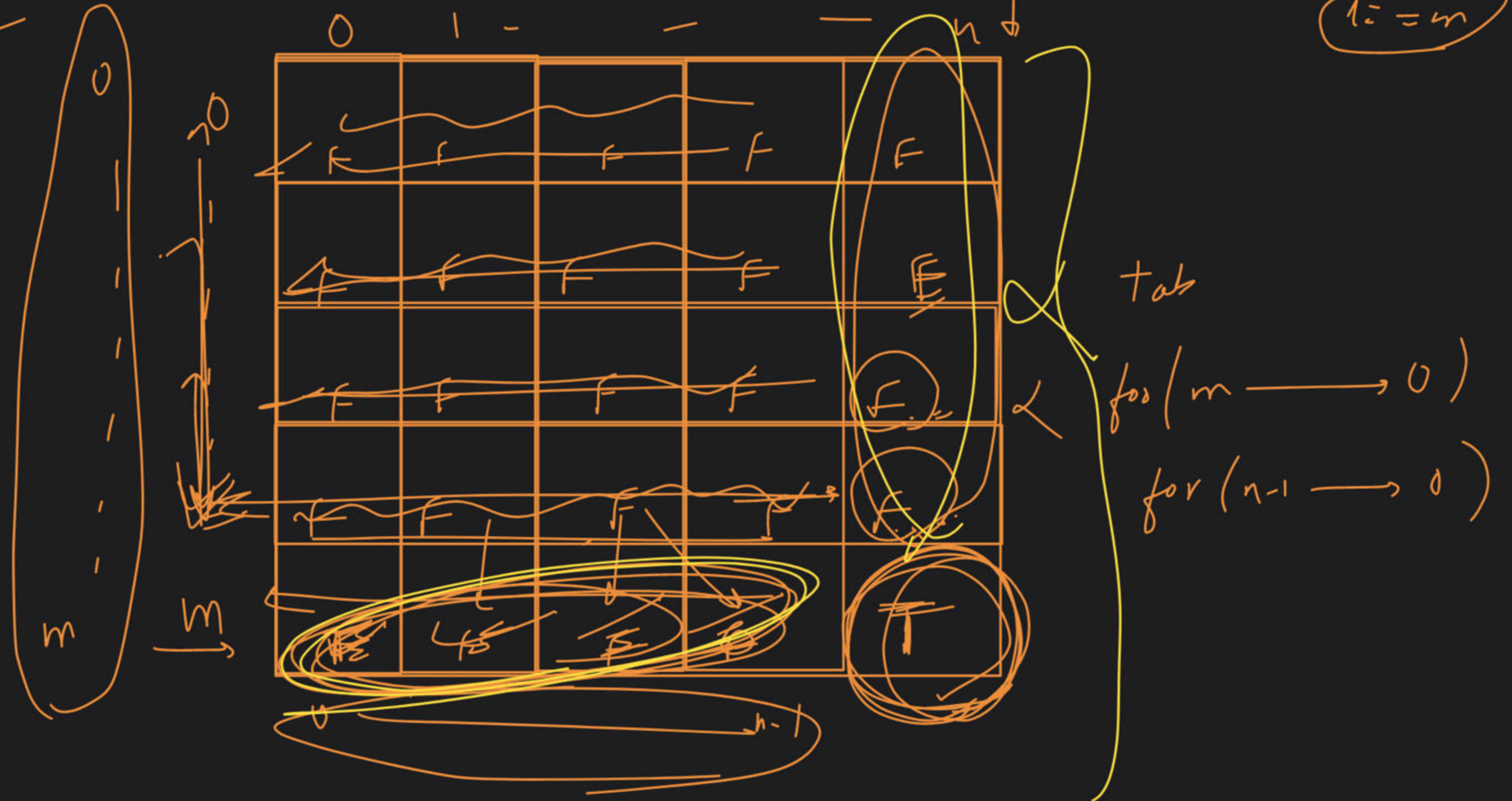
0	0	0	0	1	0
---	---	---	---	---	---

next

0	0	0	0	1
---	---	---	---	---

112m

Tabulation

$$12 = 4n$$


~~curr~~ $\rightarrow$ dp[i]

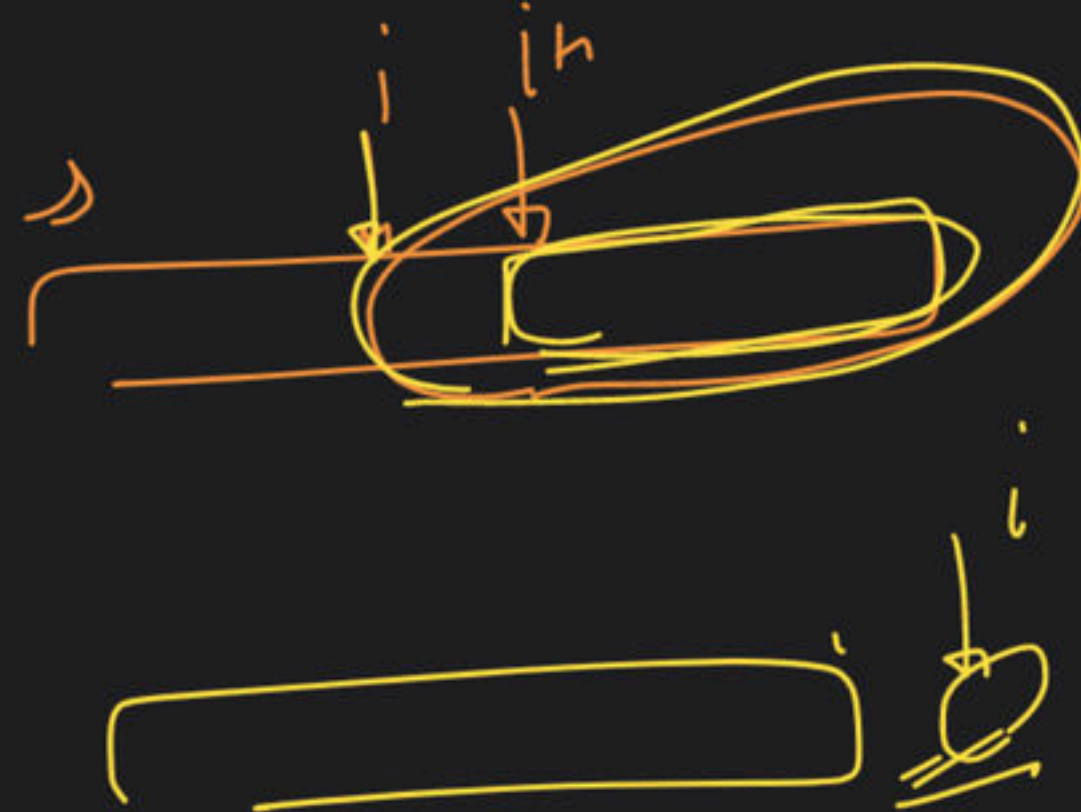
next $\Rightarrow$ dp[i+1]

curr $\rightarrow$ dp[i]

curr(n)
dp[i](n)
(curr) $\rightarrow$

dpn
 $\rightarrow$ T

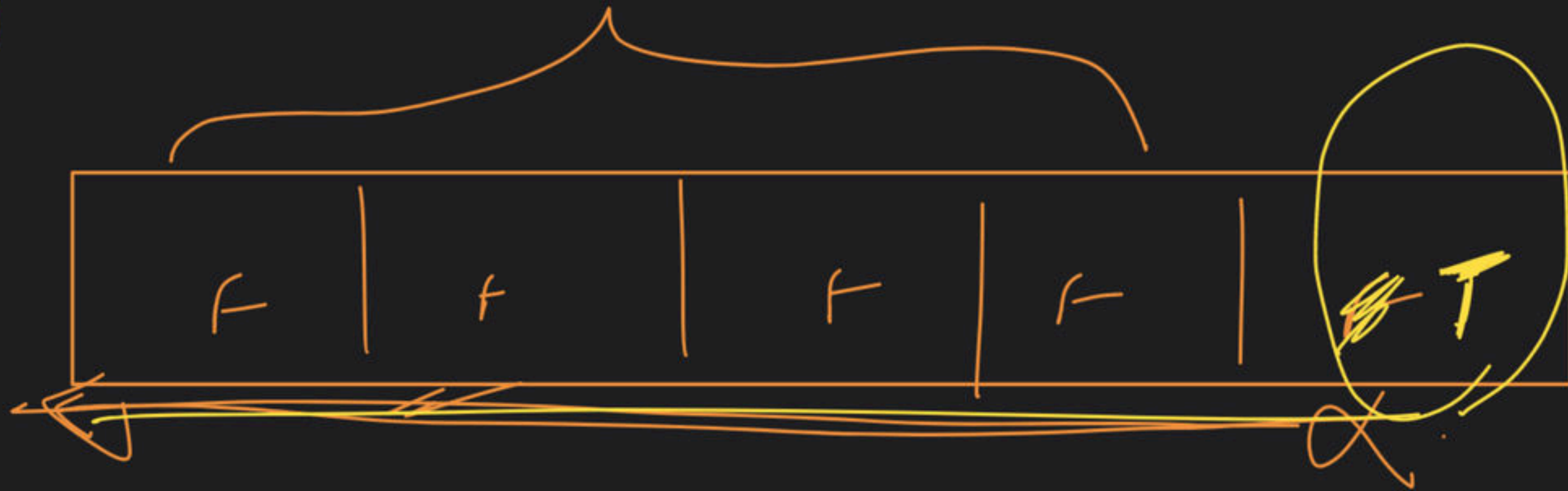
next



f(i $\rightarrow$ m $\rightarrow$ 0)
for(j $\leftarrow$ n-1 $\rightarrow$ 0)

curr $\rightarrow$ dp[i]
curr(n) = T

i = n
j = n
curr[j] = T



$\text{if } (i == n)$



$(\text{wr} \rightarrow \text{dp}[i])$

$\text{if } (i == n)$
 $\text{wr}[n] = \text{I}$

$\text{dp}[i][n]$

$\text{dp}[n][n] = \text{I}$

















